Towards Water-Sensitive Cities: Presentation Script (v4)

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What changed in v4

v4 follows the redesigned poster (v8), gives the six speakers near-equal shares, and is written to be spoken, not read. Total speaking time is about 14 minutes, including Shivang's bridge lines as anchor.

- **New poster cues.** The staircase graphic and the five-questions card are gone from the poster. Rounak now points at the "What is a water-sensitive city?" box and the three cards beneath it.
- **Three questions instead of five.** The cards ask *Can the system carry the water? Which agency is accountable, if any? Who gains, who loses, who is heard?* Rounak poses them and Saral answers them for Bengaluru, so the talk opens and closes on the same frame.
- **A hook that comes back.** We open on Majestic bus station standing on a lake bed, and close on it.
- **Balanced load.** Every speaker has 2:00 to 2:25, instead of 1:30 to 2:45 in v3.
- **Named roles.** Shivang opens and anchors (one bridge line between each part, and chairs Q&A); Rounak, Vaidurya, Soni and Saral take parts 2 to 5; Vinayak closes.

What the marking sheet rewards, and where we earn it

The HSS-304 sheet scores six areas. Each row below names the moment in the script that earns it, so everyone knows what their part is doing.

Criterion	Where the script earns it
Intellectual clarity: originality	The claim that drought and flood are one problem, and that Bengaluru went backwards from a lake city (Shivang, Vaidurya)
Intellectual clarity: use of evidence / examples	Dated decisions and named places (Vaidurya), three numbers on the poster (Soni), the tanker price cap and Madina Nagar evictions (Saral)

Criterion	Where the script earns it
Structure: clear objectives and introduction	Shivang states three aims and a roadmap, naming every speaker, inside the first two minutes
Structure: logical flow and coherence	Shivang's one-line bridges ask the question the next speaker answers; the three questions bookend Rounak's and Saral's parts
Structure: time management	Timings below; one cuttable line marked in every part
Delivery: clarity, articulation	Short sentences, numbers said slowly, pauses marked
Delivery: confidence and engagement	Open on a striking fact about a place the audience knows; every speaker points at the poster; no reading from paper
Delivery: use of academic language	Produced nature, urban political ecology, informality, selective enforcement, water balance, catchment governance: each used once, then explained in plain words
Visuals: poster design and readability	Every part names exactly what to point at
Visuals: relevance of visuals to content	Photos, timeline, numbers and both maps are each used by a speaker; nothing on the poster goes unmentioned
Scholarly context: engagement with key literature	Eight readings, each tied to a specific claim, not just name-checked
Scholarly context: accuracy and consistency of references	Author and year said the same way every time; accuracy notes at the end
Discussion: ability to respond to questions	Q&A section assigns each likely question to an owner, with a 20-second answer

Running order

Six parts of 2:00 to 2:25 each, about 1,880 words at roughly 140 words a minute, plus about 40 seconds of Shivang's bridge lines: about 14 minutes in all. If the group runs long, drop the line marked *(cut if short on time)*, never a whole beat.

Speaker	Part	Time	Words	Point at on the poster
1 · Shivang	Opening: the lake under the bus station; then anchor lines between every part	2:00 + ~40 s of bridges	~300	The title band, then the left photo, then the right photo
2 · Rounak	Framework: what a water-sensitive city is, and three ways to study it	2:20	~320	"What is a water-sensitive city?" box, then the three cards left to right
3 · Vaidurya	Case study: how the lake city was unmade	2:10	~310	The timeline, 1898 to 2024
4 · Soni	Technical explanation	2:05	~290	Bellandur photo, the three numbers, map markers 2 and 3
5 · Saral	Political and social explanation	2:25	~340	The four labelled points in the political column, map markers 1 and 4, then back up to the three cards
6 · Vinayak	Solutions and conclusion; closes the talk and invites questions	2:20	~330	Solutions band, the "Bengaluru, rebuilt" map, then the green Conclusion panel

Delivery rules for everyone

- Learn your part as beats, not sentences. Say it in your own words; only the first line, the numbers and the handover need to be exact.
- Point at the poster every time you say "here" or "this". Stand beside it, not in front of it.
- Slow down on numbers. Say "two hundred and seven, down to ninety-three", then pause.
- Look at the examiner, not the poster, when you say your section's last line.
- Say the author and the year the same way every time (for example "Ranganathan, 2015").
- The next speaker steps forward while Shivang says the bridge line, so there is no dead air.

Speaker 1, Shivang: Opening (2:00), then anchor throughout

Poster cue: stand beside the title band, facing the room. Say the first lines without looking at the poster; turn to the left photo, then the right one, only when you mention them.

Good morning, everyone.

In the heart of Bengaluru stands Majestic, the city's busiest bus station. Thousands of people pass through it every day. Almost none of them know they are standing at the bottom of a lake.

(Beat.)

Majestic is built on the bed of Dharmambudhi Lake. And that one fact is the whole story of our poster.

Look at this photo on the left. March 2024. Almost half of the city's government borewells, 6,900 out of 14,000, have run dry. Bengaluru is short of about 500 million litres of water a day, and families are paying double the legal price for a tanker.

Now look to the right. October 2024. Same city, same year, and Tatanagar is under water.

Too little water in March. Too much in October. It looks like bad luck, twice. Our argument is that it is one problem with one cause: more than a century of decisions about land, pipes and power.

(Slow down. Look at the examiner.) Bengaluru did not simply run out of water. It built over the places that used to hold it.

So we have three aims today.

1. Explain what a water-sensitive city is, and why it matters in the Anthropocene.
2. Show, through one city, how a place that once held its rain lost that ability.
3. Propose solutions that change technology, institutions and justice together.

And there are six of us to get you there. Rounak sets out what a water-sensitive city is and how we study it. Vaidurya takes you through a century of the city's lakes. Soni explains what the water physically does now. Saral asks the harder question: who decided? And Vinayak brings it home with solutions.

Shivang (anchor): "So first: what would a city that gets water right actually look like? Rounak."

Shivang's anchoring job. After each part, Shivang says the one-line bridge marked *Shivang (anchor)* and names the next speaker, who steps forward while it is said. Shivang does not close the talk; Vinayak does. In Q&A, Shivang chairs: repeat the question so the room hears it, then pass it to its owner.

Speaker 2, Rounak: Framework, three ways to study a city's water (2:20)

Poster cue: the "What is a water-sensitive city?" box. On "three ways", trace the line down to the cards and touch each one as you name it.

Most cities treat water as three separate jobs. Bring clean water in. Push dirty water out. Get the rain off the streets as fast as possible. Three pipes, three departments, three budgets.

A water-sensitive city does the opposite. It treats rain, rivers, groundwater and wastewater as one cycle, and plans the city around it. Rain is not waste to throw away; it is a resource to hold. Wong and Brown, in 2009, describe cities moving through stages: from just supplying water, to sewering it, to draining it, and finally to working with the whole cycle.

Why now? Because we live in what many scholars call the Anthropocene, an era in which humans are a force shaping the planet itself. At city scale that is very concrete. We pave catchments, fill lakes, and pump aquifers faster than the rain can refill them.

But water in a city is never just nature. Neil Smith calls it *produced nature*: already reshaped by labour and capital. And Kaika and Swyngedouw, 2000, point out that once pipes go underground, the politics goes underground with them. Water seems to arrive on its own, and nobody asks who was left unconnected.

So, as the poster shows, we study Bengaluru in three ways.

- **Technocratic:** can the system carry the water? This matters, because water follows physics. But Ranganathan warns that this view on its own hides the causes of risk.
- **Bureaucratic:** which agency is accountable, if any? Ananya Roy reminds us that the state decides when its own rules apply.
- **Political ecology:** who gains, who loses, who is heard? Hädrich Silva and colleagues, 2024, show that water-sensitive design can deepen inequality if nobody plans for that. *(cut if short on time)*

Keep those three questions in mind. We will come back and answer them.

Shivang (anchor): "Thank you, Rounak. Three questions, then. Let's put them to Bengaluru, starting in 1898. Vaidurya."

Speaker 3, Vaidurya: Case study, how the lake city was unmade (2:10)

Poster cue: the timeline. Start on 1898 and walk your hand right, one dot per date.

Our case study is Bengaluru. Before the dates, picture the city as it was. There is no big river here, so Bengaluru grew around tanks, called keres, linked by channels. When one lake filled in the monsoon, it spilled into the next one downhill: a staircase of water, storing the rain and soaking it into the ground. The region once held roughly a thousand of these lakes.

Now follow the timeline with me, left to right: four dark dots for decisions, then two coloured dots for the crises.

1898. Plague hits the city. Colonial planners recast the channels between those lakes as sanitary drains. Ranganathan shows this is the first big shift in thinking: water that used to feed the next lake becomes waste to be carried away.

1937 to 1969. Piped water arrives, and the lakes stop looking useful. Sampangi Lake is drained and becomes a stadium. The bed of Dharmambudhi Lake goes to the state bus corporation, and becomes Majestic, the bus station we started with.

1974. Cauvery river water arrives, pumped about 95 kilometres, and uphill. Now the lakes look like spare land.

1990s to 2010. The IT boom turns wetlands into real estate. Researchers at IISc counted 207 water bodies in Greater Bangalore in 1973. By 2010, there were 93. More than half, gone.

And then the two crises: the Outer Ring Road flood in 2022, and the borewells running dry in 2024.

So what does this line tell us? No single storm or drought broke the system. A chain of choices did, and each one made sense to someone at the time.

Shivang (anchor): "Thanks, Vaidurya. Which leaves a practical question: with the lakes gone, where does the rain actually go? Soni."

Speaker 4, Soni: Technical explanation (2:05)

Poster cue: the Bellandur photo; then each of the three number boxes as you say it; then markers 2 and 3 on the "Where it fails" map.

Technical explanation. Start with this photo. That is Bellandur, the city's largest lake, in 2018: so choked with weeds that it is being cleared by machine.

The old cascade did two jobs at once. In the monsoon it caught the rain and passed the overflow from lake to lake. All year round it let water seep into the ground and refill the aquifers. One system handled both flood and drought.

Cut the cascade, and both jobs fail.

Job one: floods. *(Point at 207 → 93.)* With more than half the water bodies gone and channels built over, the overflow has nowhere to go. It runs across the road instead of into the next lake. That is marker 2 on the map: the Outer Ring Road in 2022, a tech corridor built through low-lying lake and valley land. Marker 3 is Tatanagar, flooded in 2024.

Job two: drought. *(Point at 6 of 6.)* A 2024 water balance by WELL Labs found that all six groundwater units in urban Bengaluru are over-exploited. Natural recharge is only about a tenth of what the city uses every day. The ground is not refilling, while more and more borewells pump it out.

And the backup? *(Point at 500 / 2,600.)* A river about 95 kilometres away, pumped uphill. Even with it, the city was about 500 million litres a day short in 2024.

So here is the paradox in physical terms. Bengaluru throws away its own rain as flood, buys water from far away, and still runs short.

But physics cannot tell us why a lake became a bus station. Or why some neighbourhoods get pipes and others get tankers.

Shivang (anchor): "Thank you, Soni. So the water explains what failed. It doesn't explain who decided. Saral."

Speaker 5, Saral: Political and social explanation (2:25)

Poster cue: the "Political and social explanation" column; map marker 1, then marker 4; at the end, look up to the three cards at the top.

Political and social explanation. Four points on the poster, each backed by a reading.

First: lakes became land. As Kaika and Swyngedouw show, once water came through pipes, the lakes stopped looking like infrastructure. They looked like empty land, and in a booming city, empty land is worth a fortune. That is marker 1: lakes built over, right in the city centre.

Second: selective enforcement. This is Ranganathan, 2015, on storm drains. After a flood, who gets blamed? "Encroachers" on lake beds and channels. But look at who actually gets evicted. Lower-middle-class settlers in Madina Nagar, living on a filled storm channel, face demolition. A multinational's offices on wetland next door do not. Same rule: applied to the weak, waived for the powerful.

Third: unequal supply. This is Ranganathan, 2014, on water "mafias". On the city's unauthorised edges, like marker 4 in the south, only 10 to 30 percent of households in her fieldwork had piped water. Everyone else relied on borewells and private tankers. And those tanker operators are not outside the state. They work alongside officials, in the gap the state leaves.

Fourth: rules as a choice. Ananya Roy argues that the state decides when its own rules apply. In March 2024 the government capped tanker prices at 1,000 rupees for 12,000 litres, and ordered every tanker to register. A week later, people were still paying over 2,000. The rule existed. Enforcing it was a choice.

So let's answer our three questions.

- **Can the system carry the water?** Not any more, because the land it ran through was sold off.
- **Which agency is accountable?** In practice, none on its own. Lakes, drains and supply sit with different bodies, and land agencies and builders shape the outcome more than the water board does.
- **Who gains, who loses, who is heard?** Developers and tanker operators gain. Settlers on filled channels and the tanker-fed edge lose, and they are the least heard.

Shivang (anchor): "Thanks, Saral. If the lakes were lost through land, pipes and power, then the fix has to work on all three. Vinayak, take us home."

Speaker 6, Vinayak: Solutions and conclusion (2:20)

Poster cue: the Solutions band, one card per register; then the "Bengaluru, rebuilt" map; finish standing by the blue Conclusion panel.

Solutions. One answer in each register, and they only work together.

Technocratic: reconnect the lake cascades and protect the valleys they drain. Refill lakes with treated wastewater, a large source the city barely uses. And build recharge wells, so rain soaks back into the ground.

Bureaucratic: put lakes, storm drains and sewage under one catchment authority, instead of four agencies pointing at each other. Register and meter every tanker, and publish the data. And enforce the price cap, not just announce it.

Political ecology: apply encroachment rules to corporate and state projects first. Secure people's tenure before any eviction for lake works. And give lake users and the periphery a seat at the table.

This small map shows what that adds up to: the same valleys, the lakes linked again, green buffers along them, and piped water reaching the edge of the city, not just the core.

Why insist on all three? Because restoring a lake can itself become an excuse to evict the poor. That is exactly Hädrich Silva's warning. So tenure comes before construction.

We should be honest about the limits too. Hydrology is real. No lake system absorbs every storm, and a growing city will always need some imported water. We are not saying the rain is political. We are saying that who floods, and who goes dry, is. *(cut if short on time)*

(Step to the Conclusion panel.) So, to conclude. Bengaluru once held its rain. Politics, not just hydrology, broke the cascade. A water-sensitive Bengaluru must rebuild the lakes and the rules together.

David Harvey calls the right to the city the right to change ourselves by changing the city. For water, that means a say in where water goes, and whom it reaches.

So the next time you wait for a bus at Majestic, remember: you are standing in a lake. Thank you, and we'd be happy to take your questions.

Q&A: who answers what

Shivang chairs: repeat each question so the room hears it, then pass it to its owner by name. The owner answers in about 20 seconds; anyone may add one sentence. Shivang takes any question nobody owns.

Likely question	Owner	Short answer
"The brief's example is Delhi. Why Bengaluru?"	Shivang	The course is urban political ecology, and in Bengaluru politics is the cause, not the backdrop. The lakes were lost through decisions we can date, and two of our key readings are Bengaluru fieldwork.
"Was the old lake system really water-sensitive?"	Rounak	Not in the modern technical sense. But it did what the last stage asks: it held rain, passed it on and recharged the ground. So we say it was water-sensitive "once, in part".
"Is the Anthropocene official?"	Rounak	No. In March 2024 geologists voted against making it a formal epoch, but still called it a useful description of human impact. We use it in that sense.
"Why does your timeline start in 1898, not when the lakes were built?"	Vaidurya	Before 1898 the lake system is working. 1898 is the first decision that changed what the lakes were for, so that is where the story of their loss begins.
"Isn't this just population growth?"	Soni	Growth raised demand. But where the city grew, onto lake beds and channels, and who was allowed to stay there, were decisions.

Likely question	Owner	Short answer
"Why not just blame encroachers?"	Saral	We do blame encroachment, of every kind. Our point is that enforcement picks the weakest encroacher.
"Isn't one catchment authority just more bureaucracy?"	Vinayak	It replaces several bodies instead of adding one. The point is that one body answers for the whole water cycle.
"What would you do first?"	Vinayak	Freeze building on lake beds and valley zones, and publish tanker supply data. Everything else depends on those two.

Readings

Each speaker should be able to name their reading and its idea in one sentence if asked.

Reading	Who cites it	The idea you use
Wong and Brown (2009); Brown, Keath and Wong (2009)	Rounak	Stages toward the water-sensitive city
Smith, *The Production of Nature*	Rounak	Urban water is produced nature
Kaika and Swyngedouw (2000)	Rounak, Saral	Buried networks hide social relations; piped water made lakes look like land
Roy (2005), *Urban Informality*	Rounak, Saral	The state decides when its own rules apply
Ranganathan (2015), *Storm Drains as Assemblages*	Rounak, Vaidurya, Saral	Flood risk is produced; encroachment rules are enforced selectively
Ranganathan (2014), *Mafias in the Waterscape*	Saral	Tanker operators as everyday public authority
Hädrich Silva et al. (2024)	Rounak, Vinayak	Water-sensitive design must plan who gains and who loses
Harvey (2008), *The Right to the City*	Vinayak	Collective control over how the city is remade

Accuracy notes

- The 6,900 borewells and the 500 MLD shortfall are the Chief Minister's statement of March 2024, not an audited count.
- Cauvery pumping: say "about 95 km, uphill" and avoid quoting the height; sources disagree on the lift.
- Dharmambudhi Lake: the land went to the state bus corporation in 1963 and Majestic opened in 1969. Sampangi Lake was drained in 1937 and the stadium opened in 1946.
- "Roughly a thousand lakes" is the region's historic count, not what survives today.
- The poster quotes 131 mm of rain for September 2022; have an IMD or news source ready if asked.
- The maps use OpenStreetMap data. The dotted valley lines and hotspot circles are our approximations; say "roughly here".
- Never say urbanisation alone causes every flood or dry borewell; say it changes how water moves.

Sources

- Bengaluru 500 MLD shortfall and dry borewells (Deccan Herald)
- Government caps tanker prices, March 2024 (Deccan Herald)
- Despite price cap, water costs over ₹2,000 a load (The Hindu, 14 March 2024)
- WELL Labs Bengaluru water balance, summarised (Newslaundry)
- IISc wetlands of Bangalore (Ramachandra and Aithal)
- The lake that became a sports stadium (RCC Arcadia)
- The lake that became a bus terminus (RCC Arcadia)
- Cauvery water supply history (Deccan Herald)
- Photos: tanker refill point (Saiphani02, CC BY-SA 4.0); Tatanagar flood, Oct 2024 (Shyamal, CC BY-SA 4.0); Bellandur Lake, 2018 (Ashwin Kumar, CC BY-SA 2.0)
- Course readings as listed above.